

Measuring Unemployment Insurance *Generosity*: A Robustness Replication[†]

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Abstract

This paper refines the results of [Pallage et al. \[2013\]](#), focusing on the *Generosity* of unemployment insurance programs. We maintain the core structure of their method while addressing subtle discrepancies in their numerical simulations and introducing additional considerations. We conduct a three-stage computational analysis, replicating their results, refining the convergence criterion, and introducing the leisure cost of working into the numerical simulations. Our findings indicate that the *Generosity* of the program in the UK is both higher and exhibits a smoother trend than previously estimated.

Keywords: Unemployment Insurance; Social Security; Labor Market.

JEL Classifications: C32, E21, E32.

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1 Introduction

Unemployment insurance (UI) programs are complex to evaluate because they are characterized by various factors, including benefit amount, duration, asset tests, waiting time, and more. Moreover, the perceived value of these programs to workers is contingent upon the prevailing economic conditions. A more generous unemployment insurance program is particularly appreciated during periods of limited job opportunities. Additionally, these programs exhibit variability across both space and time, making it challenging to gauge their *Generosity* accurately. Comparisons between programs from different countries or tracking the evolution of a country's program over time present intricate challenges.

In an important contribution, [Pallage et al. \[2013\]](#), hereafter PSZ, introduce a methodology designed to address the intricate challenges associated with measuring and comparing the *Generosity* of unemployment insurance programs. PSZ devises a novel approach aimed at quantifying the *Generosity* of these programs using a unified metric. They construct an initial model that encapsulates all the complexities inherent in unemployment insurance policies. This model incorporates heterogeneous agents facing liquidity constraints but possessing the ability to self-insure. Subsequently, PSZ develops a second model identical to the first in most respects, with a key distinction: the unemployment insurance policy is simplified to be one-dimensional, devoid of waiting periods, eligibility limits, or asset tests, but featuring constant benefits. The objective is to determine the level of benefits in this streamlined model that renders society indifferent between the two policies. They call the parameter that adjusts as *Generosity*. Finally, PSZ applies this

measurement strategy to evaluate the unemployment insurance program of the United Kingdom.

Our study aims to refine some results presented by PSZ, address subtle nuances in their numerical simulations, and introduce additional considerations. PSZ adopt the utility framework of [Hansen and İmrohoroglu \[1992\]](#), but the expression reported in their paper differs slightly in its normalization, and the code implementation differs slightly again. We stress, however, that these differences amount only to affine transformations of period utility and therefore do not, by themselves, affect preferences or equilibrium outcomes. We retain this discussion solely for replication transparency, in order to document precisely the relationship between the model as described in the original article and the model as implemented in the original code.

A further scrutiny of PSZ’s work reveals an additional factor in their model related to the leisure cost of working. Inconsistencies in accounting for this cost in their numerical simulations have notable consequences for the model’s equilibrium. We address this by incorporating the leisure cost into our algorithm, enhancing the fidelity of our computational results.

Our computational work unfolds in three stages. Initially, we replicate PSZ’s results, revealing a slight deviation in outcomes due to the strength of the convergence criterion used in their algorithm. We then enhance the convergence criterion, observing a smoother trajectory in our *Generosity* measure. Subsequently, introducing the leisure cost of working produces a significant shift in the estimated *Generosity*, challenging PSZ’s earlier findings and emphasizing the nuanced impact of program features beyond benefit

amounts.

Comparing our *Generosity* measure over time with PSZ's, we observe a consistent decline during the 1980s and a subsequent period of stability in the 1990s. While our results align with PSZ's claim of retrenchment during the Thatcher (1979 – 1990) and the early Tony Blair years (1997 – 2002), we diverge from their assertion of a positive trend during the John Major years (1990 – 1997). In addition, we find a smoother trajectory for the *Generosity* over time than reported in PSZ.

We undertake a comparative analysis of the computed economic characteristics throughout our study. This involves contrasting the detailed and simplified models, encompassing both the novel outcomes considering the leisure cost of working and the results obtained by PSZ. The introduction of a leisure penalty for accepting a job offer leads to a reduced share of the workforce being employed. Consequently, this scenario results in a higher number of individuals receiving some benefit, necessitating an increased tax rate to maintain equilibrium in the budget of the social programs.

We show that when there is no cost associated with leisure, workers have a greater incentive to accept job offers, as the only motivation is the increase in income. As a result, the taxable base of the economy is larger, allowing the government to maintain a lower tax rate, since more people are employed and contributing to taxes. On the other hand, when the model accounts for the cost of leisure, wealthier individuals, in particular, may choose to reject job offers, as the additional financial benefit may not outweigh the loss of leisure. This reduces the taxable base of the economy, as fewer workers are accepting jobs and thus paying taxes. To compensate for this reduction in the tax base and to ensure

that the budget for programs such as unemployment insurance and other social benefits remains balanced, the government needs to increase the tax rate.

This paper is organized as follows: Section 2 presents the model proposed by PSZ. In Section 3, we discuss the specificities of our approach, comparing it to PSZ's, and additionally introduce the dataset used. Section 4 describes our computational strategy for replicating and producing more concise results. In Section 5, we present the replication of the *Generosity* estimates from PSZ, along with the results obtained under our leisure-cost implementation and other model indicators, such as taxes and unemployment. Finally, Section 6 concludes the paper.

2 Model

This study builds upon the methodology developed by PSZ, which was employed to measure the Unemployment Insurance (UI) Generosity in the United Kingdom. Specifically, the measurement of the *Generosity* involves comparing two scenarios. In the first scenario, the detailed model is considered, taking into account the specific parameters of an unemployment insurance program - for example: duration, amount, etc. A simulation of this economy is then performed, and the cross-sectional average utility of agents in the stationary distribution is calculated. In the second scenario, a simplified model is solved, where the unemployment insurance program has only one adjustment parameter, defined as *Generosity*. We follow PSZ in using average utility as the welfare criterion; alternative summaries, such as median utility or other distribution-sensitive measures, could also be considered, but are beyond the scope of the present replication exercise.

We can set a feature vector (employability rate, eligibility requirements, benefit duration, etc.) for the detailed model, while a simplified model uses the simplest form possible, with unemployment insurance being offered to the unemployed individual without any other clauses. Both models consist of a continuum of people, each characterized by their employment status, asset level, and eligibility for unemployment insurance.

Time is discrete, and each period is represented by t . The labor market is modeled as a lottery system, where there may or may not be a job offer for participants. If an offer is made, the individual decides whether to supply a fixed indivisible amount of labor, denoted by $0 < \hat{h} < 1$, or not work. Here, $\hat{h}$ represents the fraction of the normalized time endowment devoted to work, not an absolute number of hours. Employment is discrete and indivisible in each period, and the decision has a discrete impact on leisure.

Individuals care about their consumption, c_t , and leisure, l_t , which are chosen to maximize their expected discounted utility. Their utility in each period t is represented by $u(c_t, l_t)$, where u is an increasing and concave function in both arguments. Individual wealth, m_t , can be used as self-insurance against adverse shocks, but there is no possibility of borrowing, i.e., $m_t \geq 0$.

Concerning the stochastic labor market, individuals enter a lottery at the beginning of each period that will determine their employment status. The lottery outcome, k_t , for each individual, is $k_t = o$ if a job offer is extended or $k_t = n$ if it is terminated. The probability of receiving a job offer depends on whether the individual had an offer in the previous period. In any case, the agent can accept or reject the job offer. This decision is represented by x_t , where $x_t = 1$ if the job is accepted in period t , and $x_t = 0$ if it

is rejected. Each person's output is given by y , which is also their gross and constant income over time.

The savings from one period to another are described as:

$$m_{t+1} = m_t + y_t^d - c_t, \quad m_t \geq 0 \quad \forall t, \quad (1)$$

where y_t^d , the disposable income, can take one of four values:

$$y_t^d = \begin{cases} (1 - \tau)y, & \text{if employed} \\ (1 - \tau)y\omega, & \text{if unemployed and in the waiting period for UI} \\ (1 - \tau)y\theta, & \text{if unemployed and eligible for UI} \\ (1 - \tau)y\psi, & \text{if unemployed and not eligible for UI,} \end{cases}$$

with τ representing the tax rate on income necessary to finance the unemployment insurance program and other social programs, ω is the income fraction when the person loses their job and is in the waiting period to receive UI, ψ another possible social program to which the individual may be entitled when not eligible for UI, and θ represents the fraction of income received in the form of unemployment insurance. In this model, all income is taxed, including that received from the government (θ , ψ , and ω) and $\theta > \psi$.

Here and throughout the paper, "unemployed" refers to individuals who are not working in the current period. Thus, an individual who receives but rejects a job offer is treated as unemployed; following PSZ, this individual is immediately ineligible for UI benefits and receives the income associated with the ineligible-unemployed state.

In the simplified model, the person is always eligible, so the variables ω and ψ are

ineffective. In this case, *Generosity* is directly represented by the variable θ . Thus, the potential income in this model is:

$$y_t^d = \begin{cases} (1 - \tau)y, & \text{if employed} \\ (1 - \tau)y\theta, & \text{if unemployed.} \end{cases}$$

Eligibility for unemployment insurance benefits is described by several policy parameters. Following PSZ, we denote by α the vector of UI parameters, which may include the replacement rate, waiting period, benefit duration, asset tests, and other program rules. Eligibility itself depends on a vector of variables s_t . Following PSZ, we write the transition of this vector in compact form as

$$s_{t+1} = \chi(s_t),$$

where χ denotes the law of motion governing eligibility. This notation follows PSZ's compact representation of the eligibility process. In particular, as in PSZ, individuals who receive and reject a job offer are treated as ineligible for UI benefits.

The intertemporal problem of an individual is represented by:

$$\begin{aligned} [P] \quad & \max E_0 \left[\sum_{t=1}^{\infty} \beta^t u(c_t, l_t) \right] \\ & \text{subject to constraint (1),} \end{aligned}$$

where leisure is $l_t = 1$ for those not working, and $l_t = 1 - \hat{h}$ for those working (remember that work is discrete and indivisible, being equal to $\hat{h}$), and $\beta \in (0, 1)$ is the discount factor that brings future value to present value.

The individual faces two situations. The first when they have a job offer, which requires two decisions: first, whether to accept or reject the job offer; second, how to consume/save given their available income today. The Bellman equation for this situation is described as follows:

$$V(m, s, o; \alpha) = \max_x x \left\{ \max_{c, m'} u(c, 1 - \hat{h}) + \beta \int_{(k'|o)} V(m', s', k'; \alpha) d(k'|o) \right\} \\ + (1 - x) \left\{ \max_{c, m'} u(c, 1) + \beta \int_{(k'|o)} V(m', s', k'; \alpha) d(k'|o) \right\}$$

$$\text{subject to} \quad m' = m + y^d - c, \\ m' \geq 0, \\ s' = \chi(s).$$

Similarly, the Bellman equation when the individual does not have a job offer is:

$$V(m, s, n; \alpha) = \max_{c, m'} u(c, 1) + \beta \int_{(k'|n)} V(m', s', k'; \alpha) d(k'|n)$$

$$\text{subject to} \quad m' = m + y^d - c, \\ m' \geq 0, \\ s' = \chi(s).$$

After finding the value function resulting from the above maximization, $V(\cdot)$, we can use the optimal decisions for work, $x(m, s, k)$, consumption, $c(m, s, k)$, and savings, $m'(m, s, k)$, to construct the distribution of agents $f(m, s, k)$, which provides, for each state (m, s, k) , the proportion of agents characterized by that particular state.

2.1 Equilibrium

The steady-state equilibrium, given by a specific unemployment insurance policy α , is determined by choices of work, assets, and consumption for all individuals, a value function V , an agent distribution f , and a tax rate τ , such that:

1. Agents solve their individual intertemporal problems, given (α, τ) and labor market characteristics;
2. The unemployment insurance agency balances its budget;
3. The distribution of agents is invariant.

2.2 Detailed Unemployment Insurance Program

At this point, the actual unemployment insurance program is described in as much detail as possible and allowed by computational limitations. The following components are considered:

1. A waiting period a , during which the worker is not yet considered eligible to receive program benefits if they become unemployed.
2. The duration of eligibility z ; that is, the time the individual will be eligible to receive unemployment insurance benefits.
3. The proportion of the previous income that unemployed individuals will receive as a benefit, $\theta(j)$, which may vary during the unemployment period ($j = 1, \dots, z$).¹

¹In our implementation, following PSZ, the replacement rate θ is constant within each calendar-year simulation. The value of θ may vary across calendar years according to the PSZ data, but within a given year the replacement fraction is fixed.

4. The proportion of income that unemployed individuals will receive after losing their eligibility, ψ , for example, through other social programs.

Therefore, the vector of parameters to be calibrated is:

$$\alpha = (a, z, \{\theta(j)\}_{j=1, \dots, z}, \psi).$$

Or, more simply for our specific case:

$$\alpha = (a, z, \theta, \psi).$$

2.3 Simplified Unemployment Insurance Program

For the simplified model, the following will be considered:

1. The individual will be eligible as soon as they lose their job, with no waiting time, i.e., $a = 0$.
2. Eligibility will have infinite duration, $z = \infty$.
3. *Generosity*, or the proportion of income the individual will receive through UI, θ
4. Since there will be no other social programs, $\psi = 0$.

Therefore, the simplified unemployment insurance program will have the following parameter vector, which we will map to the detailed UI program:

$$\alpha = (0, \infty, \theta, 0)$$

Or, in a summarized form:

$$\alpha = (\theta).$$

The parameter θ represents the *Generosity* that is sought to be assigned to the studied unemployment insurance program. This parameter is adjusted so that the average utility equals between the simplified and detailed unemployment insurance models.

3 Data and model adjustments

In our study, we maintain the foundational structure of the model introduced by PSZ. However, during our analysis, we observed a nuanced feature in their numerical simulations that diverges slightly from the model outlined in their paper. PSZ adopted the utility function framework from [Hansen and İmrohoroglu \[1992\]](#), yet we identified a subtle discrepancy between their approach and the one presented by [Hansen and İmrohoroglu \[1992\]](#).

While [Hansen and İmrohoroglu \[1992\]](#) define the utility function using a standard constant relative risk aversion (CRRA) formulation as follows:

$$u(c_t, l_t) = \frac{(c_t^{1-\sigma} l_t^\sigma)^{1-\rho} - 1}{1-\rho}, \quad (2)$$

PSZ, in their paper, express a slightly modified function:

$$u(c_t, l_t) = \frac{(c_t^{1-\sigma} l_t^\sigma)^{1-\rho}}{1-\rho} - 1. \quad (3)$$

We use equation (2) in our simulations. It is crucial to note that the subtraction of 1 in

different parts of the function serves as a normalization step and likely does not impact the overarching conclusions drawn in their paper. However, it came to our attention that this normalization step is not consistently applied in their code. Consequently, the simulations in their study consider the following utility function:

$$u(c_t, l_t) = \frac{(c_t^{1-\sigma} l_t^\sigma)^{1-\rho}}{1-\rho}. \quad (4)$$

Since this discrepancy is only an affine normalization, it is not quantitatively consequential in itself; we report it only to improve the transparency of the replication exercise.

Upon a more comprehensive examination of PSZ's work, we identified an additional aspect of their code that warrants consideration. In their model, individuals who accept job offers experience a reduction in leisure time. Specifically, labor is represented by a constant $\hat{h}$, which represents the fraction of the normalized time endowment devoted to work. Consequently, this duration should be subtracted from the total leisure time, which is normalized to 1 in the study, from workers who accept a job offer. However, we observed that this leisure cost of working is not consistently taken into account in their numerical simulations, leading to significant implications for the model's equilibrium.

Remaining consistent with PSZ, we maintain other model parameters. Worker production is standardized to $y = 1$, and we set the weekly discount factor to $\beta = 0.999165$, corresponding to an annual discount rate of about 4.0% over a year. Additional parameter values include $\sigma = 0.67$, and $\rho = 2.5$. In terms of work, it is assumed that individuals have a weekly time endowment of 98 hours, of which an average workweek of 44 hours is supplied to work, so $\hat{h} = 44/98 \approx 0.45$. The dataset used was sourced from the PSZ

paper, providing the α vector for the years 1971 to 2002 in the UK.²

4 Computational Strategy

Our computational work is divided into three stages. Firstly, we developed an algorithm to replicate the results of PSZ. Next, we modified the convergence criterion of the value functions in the original algorithm of PSZ. In particular, we use the convergence tolerance of 10^{-5} in our simulations, while PSZ used 10^{-3} in some parts of their algorithm.³ Finally, we incorporated the leisure cost of working into the algorithm as presented in the previous section.

The computational challenge of this work lies in finding the equilibrium of the models. To achieve this, we first solved the model using the original PSZ grid, from 0 to 62.5. In those simulations, the stationary distribution placed negligible mass above 20. We therefore used a truncated grid from 0 to 20 in the leisure-cost simulations to reduce computational burden. As a robustness check, we verified that the main results are unchanged when using the larger grid.

In our more complex model, four possible states are defined for the individual: employed, unemployed in the waiting period to receive UI, unemployed receiving UI, and unemployed receiving other social benefits. These states result from combinations of eligibility (s), job offer (k), and employment decision (x). Thus, an individual cannot be in a situation different from these proposed states. In our simple model, there are only two

²The PSZ dataset can be accessed via the following link: <https://doi.org/10.7910/DVN/MVSBRO>.

³In fact, when computing the equilibrium in the model with the detailed UI program, PSZ used a convergence criterion equal to 10^{-4} , but they used 10^{-3} for simulations with the simplified UI program.

possible states: employed and unemployed receiving UI.

There is also a distribution matrix of individuals, where the rows represent the wealth grid, and the columns represent possible states. Thus, we can allocate each person to a wealth and a possible state. The sum of the values in this matrix (the mass of people) must be equal to 1.

For the first iteration, individuals are randomly distributed in this matrix. For this distribution matrix, there exists an income tax rate τ that will balance the economy. The detailed economy is solved for the given values of α , including θ . The Bellman equation is solved, and we obtain the value function and the policy function from it. Now, with the policy function defined, it is possible to update the distribution matrix of individuals, knowing, for each individual defined by wealth and current state, what wealth they will have in the future, and in which state (through the transition probability matrix). With this updated distribution matrix, a new τ is defined to balance receipts and spending in the UI and other social benefits.

With a new τ , the solution of the Bellman equation changes, as well as the choices of individuals. So a new solution to the Bellman equation is computed. This loop is repeated until the value function and the calculated τ are close enough to the values from the previous iteration, using the convergence criterion mentioned above.

For the simplified economy, the same procedure is used, but with an additional step. The θ previously defined by the vector α is now also a tuning parameter. Therefore, we start with an arbitrary value of θ , and the economy is calculated. The average of the Value function values is then calculated, weighted by the distribution of individuals

given by each asset and state. This average is compared to the average of the detailed economy. If the values are not close enough, θ is updated using the bisection technique, and the economy's equilibrium is calculated again. These iterations are repeated until the weighted average of the value function for the two economies is close enough. Thus, the θ calculated in the simplified economy is what we call the *Generosity* of the UI program.

5 Results

5.1 Replicating the PSZ main results

The initial phase of our study is dedicated to reproducing the outcomes outlined by PSZ. To achieve this objective, we crafted our algorithm, incorporating the utility function specified in equation (4). Furthermore, we established a constant value for leisure, setting $l_t = 1$ across all potential states of workers. Additionally, we adopted the identical convergence criterion employed by PSZ.

Figure 1 compares our replication of the *Generosity* with the original results from PSZ. Using the same convergence criterion as the original paper, we notice that the initial points converge perfectly, but from 1980 onwards, the results deviate slightly, yet remain close to the original ones. This deviation occurs because, in PSZ's algorithm, the convergence criterion used was not strong enough. Depending on the initial guess of the value function, the algorithm might converge to a different point.

Therefore, although we conclude that the algorithm satisfactorily replicates the results of the paper, we decided to proceed with a second stage in this research: choosing a stronger convergence criterion. Initially, the program considered that it had found the

equilibrium of the model when the error between two iterations was 10^{-3} . Now, we set this criterion to 10^{-5} . In this exercise represented by the dark grey line, we observe that the trend of the curve's movement remains the same as the original, but its motion becomes smoother. In addition, most of the points of the new exercises are higher than the original estimated *Generosity*. We also highlight that this more rigorous convergence criterion is later employed in all simulations.

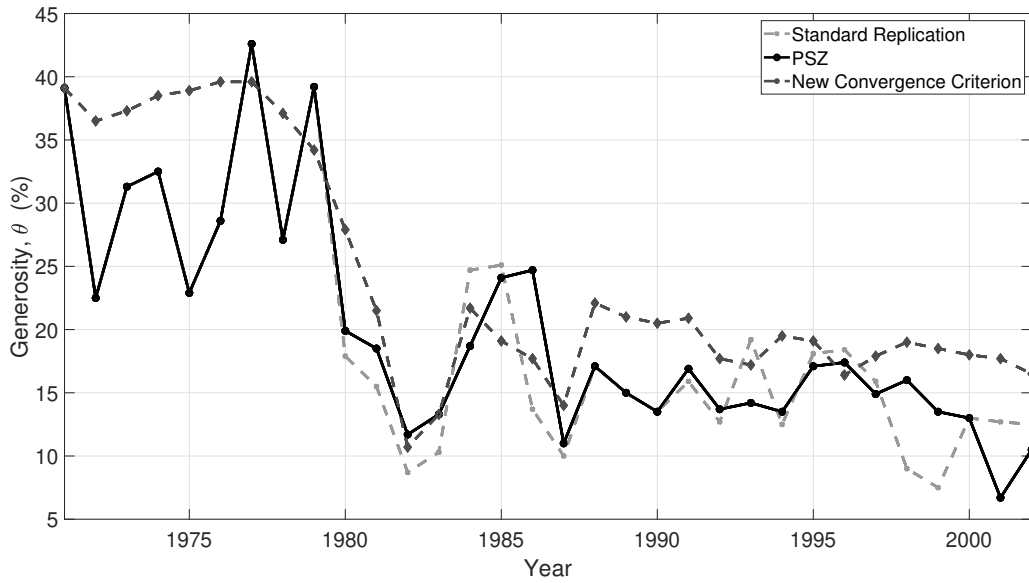


Figure 1: *Generosity: Replication versus PSZ*

Note: The dashed light gray line represents the estimated *Generosity* using the utility function from equation (4), with $l_t = 1$ and a convergence criterion of 10^{-3} . The dashed dark gray line maintains the same conditions but uses a stricter convergence criterion of 10^{-5} . Finally, the black line shows the result obtained by PSZ (see Pallage et al. [2013] page 536).

5.2 Reestimating the *Generosity* of the UI program in the UK

Next, we incorporate the leisure cost of accepting a job directly into the utility function.

Figure 2 reports three series: the black lines replicate the results presented by PSZ (Figure 5); the grey line shows the results obtained with our algorithm; and the dark-grey dashed

line represents the observed *Generosity* data.

For 1971–1981, our algorithm did not locate a value of θ in the simplified model that satisfies the welfare-matching criterion within the admissible search region and under our convergence tolerance. We therefore do not report these years. The numerical diagnostics suggest that, at high benefit levels, the interaction between higher required replacement rates, higher taxes, weaker work incentives, and a smaller tax base may prevent the algorithm from reaching a stable fixed point.⁴

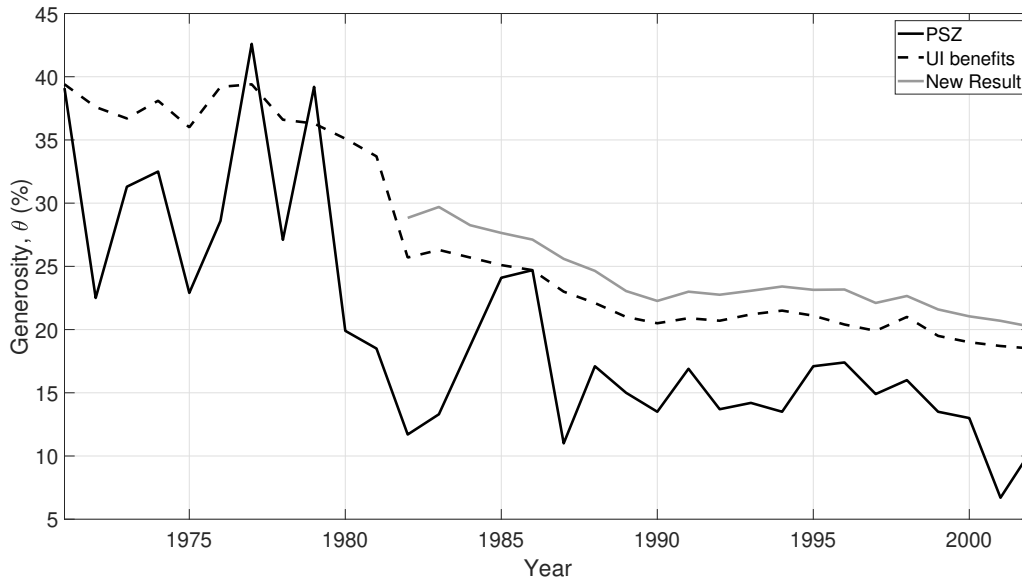


Figure 2: New estimated *Generosity* versus PSZ results

Note: The light gray line represents the estimated *Generosity* considering the costs of leisure, based on the utility function in equation (4), and a convergence criterion of 10^{-5} . For 1971–1981, our algorithm did not converge. The dark-grey dashed line corresponds to the observed *Generosity* data. Finally, the black line illustrates the result obtained by PSZ (see [Pallage et al. \[2013\]](#) page 536).

When we include leisure costs, we notice a significant change in the estimated *Generosity*. In PSZ, *Generosity* was below the government’s offered benefit, indicating that people accepted less income to streamline the system. Now, however, the estimated *Gen-*

⁴We do not interpret this numerical result as formal proof of non-existence of equilibrium.

erosity is always above the government's benefit line. This gap reflects the fact that the welfare-equivalent replacement rate in the simplified model summarizes the entire institutional package of the detailed program, including waiting periods, eligibility duration, and access to other benefits.

In any case, the outcomes derived from our newly estimated *Generosity* measure align with a decline in this indicator during the 1980s and a subsequent period of stability in the 1990s, mirroring the original findings of PSZ. However, our updated estimation presents a smoother trajectory for *Generosity* over time. Notably, we observe a negative trend persisting throughout the entire period. Consequently, our results support PSZ's assertion that the Thatcher years and the early Blair years, 1997–2002, were characterized by a retrenchment in *Generosity*. Nevertheless, our findings diverge from their claim of a positive trend during the John Major years.

5.3 *Generosity, taxation, and employment*

A crucial aspect of the model is the tax rate, which serves to balance the budget of the UI and other benefits programs. Consequently, we delve into the correlation between taxation and several significant variables in our economy. The subsequent analysis investigates the interplay among *Generosity*, taxation, and employment. It is essential to emphasize that our focus is solely on the years from 1982 onward, as these are the years during which the algorithm converged, producing interpretable results.

Figure 3 compares the tax rate τ and the (exogenous, statutory) replacement rate θ used in the detailed model. The figure shows that the tax rate responds strongly to

changes in the replacement rate. Between 1982 and 1983, an increase in the replacement rate is accompanied by an increase in the tax rate. From 1983 to 1990, the replacement rate declines, and the tax rate decreases even more sharply. A similar comovement is observed throughout the rest of the period.

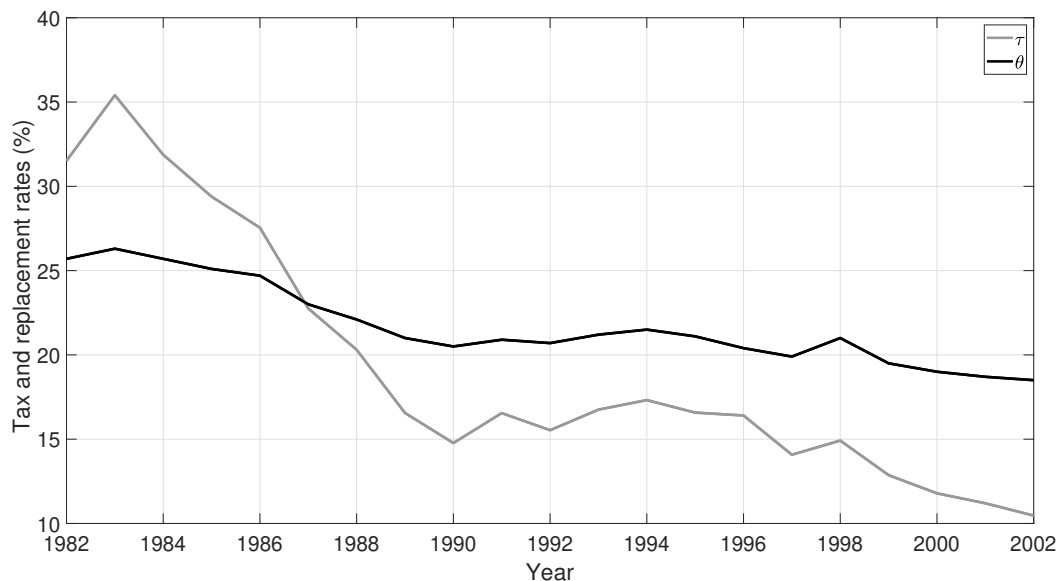


Figure 3: Taxation and replacement rate

In Figure 4, we compare the tax τ with the total number of employed people. These two values draw attention as they respond inversely, almost mirrored; that is, when τ increases, the number of workers decreases almost in the same proportion. The black line represents the absolute average between the values of τ and the number of employees, clearly illustrating the mirrored functions, with the average forming an almost straight line centered between them.

The combination of Figures 3 and 4 provides an understanding of why the model becomes unstable between the years 1971 and 1981. During these years, the government's benefits were high. With such high values, the tax τ tends to be even higher, resulting in

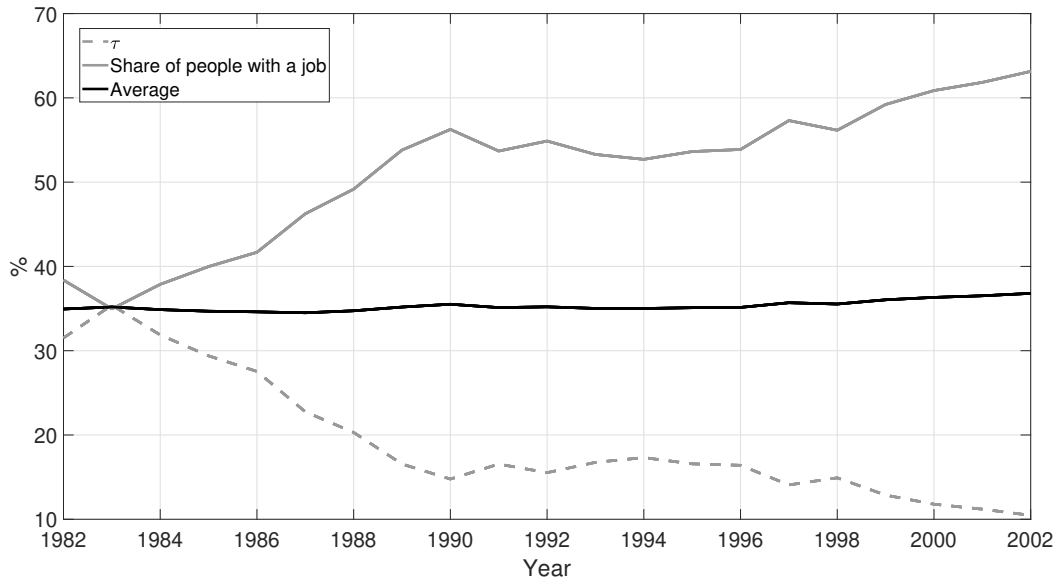


Figure 4: Taxation and employment

two effects: people will be encouraged to be unemployed and receive social benefits, and people will be discouraged from working since they will receive less for it (given the high tax). Thus, unemployment increases, reaching levels that make the economy unsustainable.

5.4 Features of the models with and without leisure cost of working

In this section, we draw comparisons regarding the characteristics of the economy computed throughout the study. The subsequent figures present a juxtaposition between detailed and simplified models, encompassing both new results (involving the leisure cost of working) and PSZ results. Since the data analyzed in this section is not available in PSZ's study, we use the values computed by our algorithm when replicating their work while keeping their convergence criterion. It is important to emphasize that labor supply is already endogenous at the extensive margin in PSZ: individuals who receive a

job offer decide whether to accept it or not. Therefore, the exercise in this section is not to introduce a new endogenous-labor extension or to recalibrate the labor-disutility parameter. Instead, our purpose is to compare the outcomes generated by the original PSZ implementation with those obtained when the leisure cost of working, which is part of the model's description and calibration, is consistently incorporated into the numerical solution.

Figure 5 shows the equilibrium tax rate τ across the four model variants. Notably, in the absence of considerations for the leisure cost of working, taxation levels are significantly lower compared to scenarios where this factor is integrated. This disparity arises because, in the absence of leisure costs, workers have a greater incentive to accept job offers, as there is no associated cost, only the allure of increased income. Conversely, when a reduction in leisure accompanies the acceptance of job offers, wealthier workers may opt to reject offers, thereby diminishing the taxable base of the economy. Consequently, a higher tax rate becomes necessary to maintain equilibrium in the budget of the UI program (and other associated benefits, in the case of the detailed model).

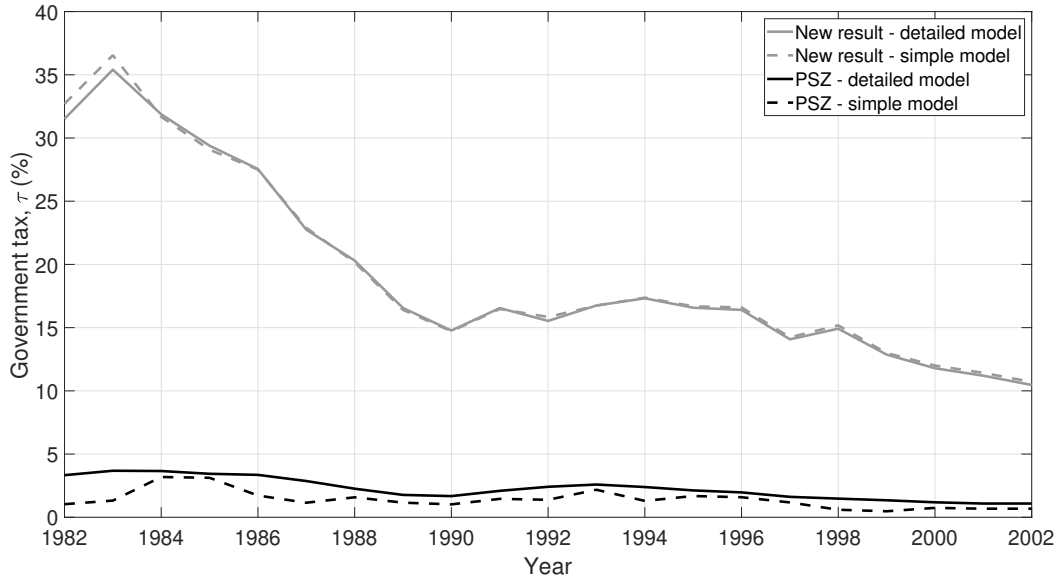
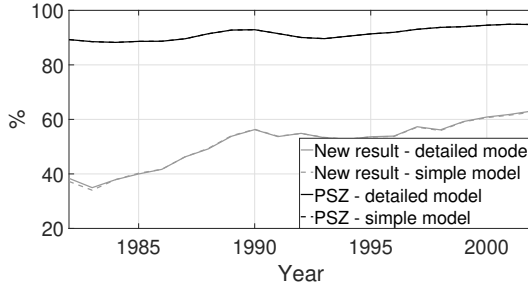


Figure 5: Government Taxation, τ

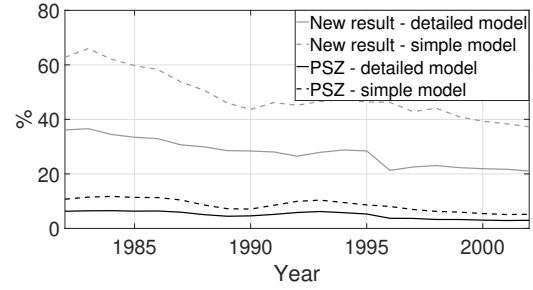
In Figure 6, we present the distribution of individuals across the four states of the model. Figure 6a illustrates that the economy exhibits a higher rate of employment in the model without considering leisure costs compared to our revised version, as anticipated. On the other hand, Figure 6b indicates a larger proportion of individuals benefiting from the UI in the model that accounts for the leisure cost of working.

Moreover, Figure 6b underscores a more frequent utilization of the UI by individuals in the simplified model. This aligns with the concept that individuals necessitate a higher threshold, θ , to be indifferent from the detailed model to a simplified one in terms of average expected utility. As a higher θ leads individuals to turn down more job offers, it is logical to observe a heightened proportion of individuals benefiting from the UI in the simplified model compared to the detailed counterpart.

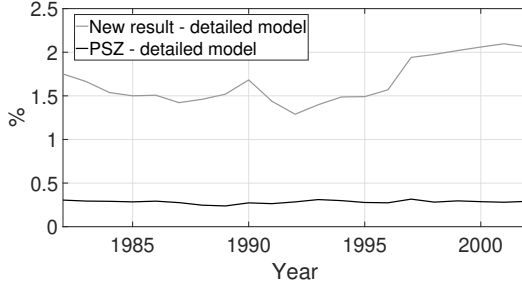
However, it is worth noting that, in the new results, the unemployment rate appears



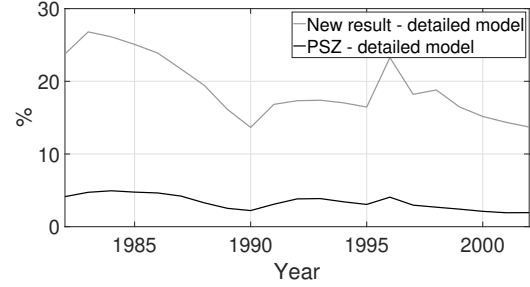
(a) Employed workers



(b) Unemployed workers enjoying UI



(c) Unemployed workers waiting for UI



(d) Unemployed workers receiving other benefits

Figure 6: Share of workers in each possible state

unreasonably high. Despite this, across all simulations, a consistent negative trend in the unemployment rate is evident throughout the analyzed period. This aligns with the decline in the unemployment rate during the 1980s but contrasts with the inverted-U shape observed in the rate during the 1990s.

Figures 6c and 6d further confirm this trend, demonstrating an increased number of individuals awaiting UI benefits and a higher count of unemployed individuals receiving other social benefits in the model that considers leisure costs.

The lower employment rate in the leisure-cost implementation arises because accepting a job offer now involves both an income gain and a loss of leisure, from 1 to $1 - \hat{h}$. As a result, some agents, especially those with enough assets to self-insure, reject job offers they would otherwise accept. This reduces the taxable base, raises the equilibrium

tax rate needed to finance benefits, and further weakens work incentives. Since we keep PSZ's original calibration fixed, this result should be interpreted as a diagnostic implication of the leisure-cost implementation rather than as a fully recalibrated alternative benchmark. A recalibration of the labor-disutility parameter or of the indivisible labor requirement would likely attenuate this mechanism.

Figure 7 highlights a notable aspect of our novel findings. In contrast to the PSZ replication where no worker turns down a job offer, our results reveal instances where workers reject such offers, particularly when factoring in the leisure cost. This phenomenon can be attributed to the comparison workers make between receiving UI and opting for a higher income, albeit at the expense of reduced leisure time.

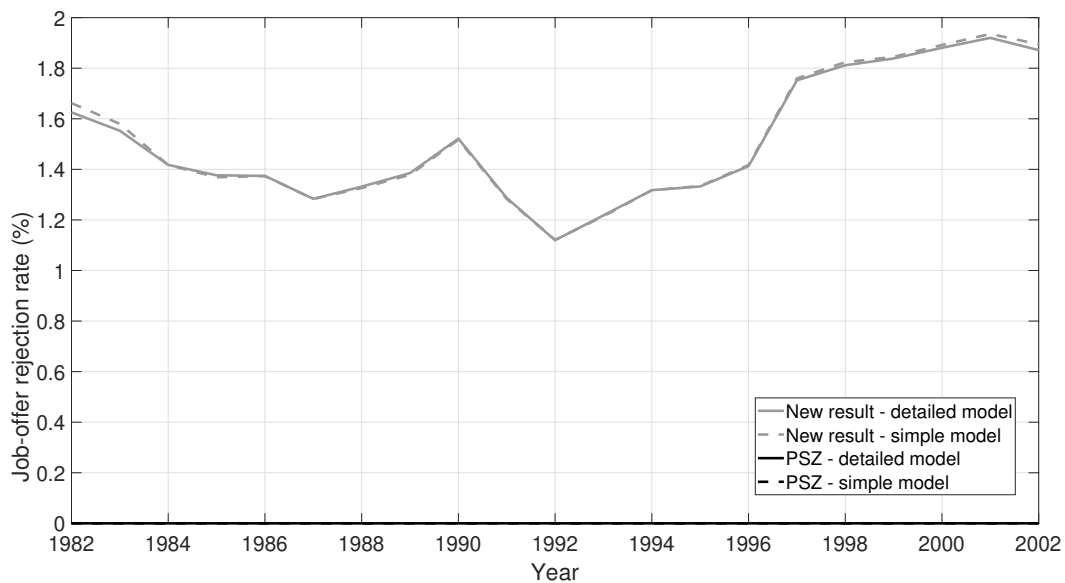


Figure 7: Share of people rejecting job offers

6 Final Remarks

In conclusion, [Pallage et al. \[2013\]](#) have made a noteworthy contribution by establishing a comprehensive model for assessing the *Generosity* of unemployment insurance programs. Despite some shortcomings in their numerical simulations, which we have meticulously addressed in this study, their work lays a solid foundation for understanding the dynamics of these programs. PSZ’s model provides valuable insights into the impact of various program features on economic outcomes, setting the stage for further exploration and refinement.

Building on the groundwork established by PSZ, our analysis contributes to the literature by reviewing, replicating, and refining the methodology proposed to measure the *Generosity* of unemployment insurance (UI) programs. The main contribution of the paper is the detailed analysis of the nuances in the modeling and calculations presented by PSZ, bringing methodological and computational adjustments that result in more accurate and robust results.

We initially identified discrepancies in the utility function used by PSZ, correcting the omission of a normalization. In addition, we consistently implement the cost of leisure term already present in PSZ’s model description and calibration but not consistently incorporated in their numerical simulations. In the model algorithm, we improved the convergence criterion, ensuring greater stability and smoothness in the trajectories of *Generosity*. In the temporal analysis, we reviewed the PSZ estimates and identified important differences, such as the absence of a positive trend during John Major’s govern-

ment, which contradicts the original conclusions. Finally, we explore the fiscal and labor market implications of including the cost of leisure, showing how it reduces the tax base, raises the tax rate needed to balance the budget, and alters the redistributive dynamics of unemployment insurance programs.

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